# FEELER REFLECTION GAME

# A CASE STUDY ON A DESIGN GAME FOR PARTICIPATORY DESIGN

Eva Durall, Teemu Leinonen
Aalto University School of Arts, Design and Architecture
✉ eva.durall@aalto.fi, teemu.leinonen@aalto.fi
Juan Fernando González
University of Barcelona, Faculty of Fine Arts
✉ gonzalezgiraldo@ub.edu

ABSTRACT

The paper explores the potential of design games as a method for developing empathetic understandings between designers and end users. With this aim, we have developed the *Feeler Reflection Game*, a board game designed to serve the contextual inquiry and early participatory design sessions of a design research looking for solutions to help people reflect on their well-being and learning. The text includes a description of the game design process, as well as an analysis of the game's use in a workshop. The case study confirms findings of earlier studies demonstrating the power of storytelling, personas, and scenarios in design. Furthermore, the case study shows that by playing the design game, the participants were able to identify themselves with the concept design and with the main situations of use. We conclude that the game acted as a boundary object that allowed participants and designers to engage in an empathic and productive dialogue.

KEYWORDS: *Design Games, Participatory Design, Empathic Design, Design Methods, User-centered Design*

## INTRODUCTION

In Participatory Design (PD) tradition, people who are expected to use the product or service have a special role. First of all, because they are experts in the field in which designers want to approach; and secondly, because in the end, the users are the ones who will benefit or suffer from the final design. For this reason, designers must ensure that the solutions provided are truly beneficial and sustainable. In order to come up with innovations that respond to the users' needs and expectations in a certain context, it is necessary to incorporate the users' insights into the design process. The Scandinavian tradition has strongly advocated democratic processes within the design project through the inclusion of points of views, as well as by offering participants the means to understand the design process and have a true impact on it. According to the Scandinavian tradition, users have the right to intervene in the design process.

PD has had an influence in areas such as HCI, user-centered design, interaction design, and information systems development. Key background assumptions and methods of the PD approach, such as the right of people to take part in decisions dealing with their living and working conditions, the awareness of the benefits that users' insights can bring to the design process, as well as the value of techniques like mock-ups, low-fi prototypes, workshops...etc., have been widely adopted (Bannon & Ehn, 2012).

In design, especially in user-centered design, empathy has been stressed as a way to achieve a better understanding of the user and the user's experience. Through empathic design (Koskinen et al., 2003), designers try to get closer to the lives and experiences of end users, in order to design a product or service that truly meets the users' needs. The PD tradition has emphasized mutuality and reciprocity (e.g. Muller, 2003) as means to improve communication and collaboration. The recognition of everyone's knowledge and the possibility to learn from each other are some of the traits that Muller attributes to hybrid spaces, situations that enable new relationships and understandings (2003). It is interesting to consider how different methods in PD, such as workshops, stories, photos, dramas, and games, among others, contribute to this hybridity. Games seem to be a promising option for developing shared

ownership, questioning assumptions, achieving mutual learning, synthetizing new ideas, negotiating, and co-creating. In PD, playing games with the end users may improve communication and provide not only new information and space for participants to contribute to the design, but also a shared space for emphatic, communal, and situated learning.

To study the potential of games in PD, we have designed several board games for various design cases. The games are custom made, since the idea is to design a game for each specific design case. In this paper, we present the *Feeler Reflection Game*, a board game designed to serve the contextual inquiry and early participatory design sessions of a design research looking for solutions to help people reflect on their well-being and learning. We describe the experience of designing the game and analyze the use of the game as a participatory practice that enables researchers and designers to develop an empathetic understanding of their participants.

In the following section, we present design games as a research technique for supporting empathy in the design process. We continue describing and analyzing the *Feeler Reflection Game*, a case in which a design game was used with the aim of augmenting understanding about the users' contexts, needs, and expectations. In the discussion, we defend the key role of design games for creating a playful and relaxed atmosphere that facilitates communication and collaboration. We conclude by considering design games as boundary objects that facilitate an empathic understanding between users and designers.

## DEVELOPING EMPATHY THROUGH DESIGN GAMES

In order to promote empathy during the design process, a variety of methods and tools have been developed. The Kouprie and Visser (2009) classification recognizes three main classes of techniques: (1) for research, (2) for communication, and (3) for ideation. Research techniques that aim to foster empathy are often focused on promoting direct contact between designers and users. In the PD tradition, Johansson (2005) has noted that 'User involvement enabled the development of a special kind of field studies that is much more design-oriented than traditional ethnographic studies' (2005, p.15-16). Hence, knowledge about the end users and the design context goes beyond descriptive practices, since researchers and designers seek to achieve an empathetic understanding of users' experiences. Mattelmäki (2006) points out the need for creative and collaborative methods that enable designers to develop open interpretations about the design context that will help them to develop empathy. The creative and collaborative practices can be, for instance, the use of *probes* (Gaver et al., 1999; Mattelmäki, 2006), *mock-ups* and *prototypes* (Ehn, 1988), *Make Tools* (Sanders & Dandavate, 1999), *design games* (Ehn & Sjögren, 1991; Johansson, 2005; Brandt & Messeter, 2004), *scenarios* (Binder, 1999), or *drama* (Brandt and Grunnet, 2000).

Some researchers see a lot of similarities between design and games. These are, for instance, their social component, evolving, and rule-based nature. According to Salen and Zimmerman (2004, p.80), 'A game is a system in which players engage in an artificial conflict, defined by rules, that results in a quantifiable outcome'. In the case of design, the design problems, the available resources, participants and their roles, as well as the work processes form boundaries for the design work and define the design space in a similar way as the rules in a game define what one can do in it. According to Vaajakallio (2012), games have been used in design for design research (Habraken & Gross, 1988), for building design competences among design students (Iversen & Buur, 2002), for empowering users, like in the 'layout kit' developed by Ehn and Sjögren (1991), and for engaging multiple stakeholders (Brandt & Messeter, 2004). Design games can be tools to define the design space. According to Botero (2013, p.59), a design space can be understood as 'the space of potentials that the available circumstances afford for the emergence of new designs at multiple levels'. Building on this definition, playing a design game can be a tool to intensify the relations among people, things, and activities. When playing a game, end users, designers, and stakeholders in general, can explore different alternatives and make collective design decisions. This transforms and expands the initially given design space by increasing empathy among participants.

Including empathic thinking in design can be a way to find inspiration and creativity about future products or design solutions (Steen et al., 2007); in other words, to broaden the design space. In PD, design games have gained popularity as a method for designers and users to jointly explore design issues on a conceptual level — to involve users in creative activity (Brandt, 2006). From this angle, explorative design games support formatting design dialogues that 'engage intended users, various stakeholders and the design team in joint inquiry into existing practice and participatory design of possible futures' (Brandt, 2006, p.61). Exploratory design games are a specific type of design game named by Brandt and Messeter (2004) that place great emphasis on engaging multiple stakeholders (Vaajakallio, 2012). Brandt (2006) distinguishes four types of exploratory design games: (1) games to conceptualize designing, (2) exchange perspective games, (3) negotiation and work-flow-oriented design games, and (4) scenario-oriented design games.

The capacity of games for creating stories, in which participants step in and play a role, helps to dissolve hierarchies, which in turn promotes a more relaxed participation. In this regard, Kronqvist et al. (2012, p.4) highlight 'Games introduce a way for engaging the participants in a mode of storytelling through the use of certain rules, material props, and visual aesthetics that set the context'. The use of storytelling in games enables players to develop an empathic connection with the situations presented. As Costikyan (2000, p.51) claims, players create experiences when participating in a game, and these generate emotional responses. 'And because a player is involved in the creation of the experience, because the experience of play is at least as much his product as that of the game designer, the emotions he feels can affect him much more deeply than the surface, empathic response you feel when viewing or reading


http://de2014.uniandes.edu.co | October, 2014. ISBN 978-958-774-070-7. pp. 349-356

about characters in a story'. Design games can take advantage of storytelling as a strategy to produce empathy among participants and, therefore, allow for a better and deeper understanding of the design concept and its challenges.

From the design perspective, storytelling has proved useful for designers. The use of techniques such as the creation of personas, scenarios, storyboards, and role-playing has been applied to help designers gain understanding and empathy towards users' experiences (Buchenau & Suri, 2000). As Fulton Suri states (2003, p.52), empathic design aims at 'achieving greater awareness, an extended imagination and sensitivity to another person's world in a powerfully memorable way'. Building on these contributions, we can claim that games create an enjoyable atmosphere that allows equal communication and facilitates sharing experiences between participants. Smooth communication and willingness to collaborate are key conditions for understanding how people experience things and for developing empathy with their world visions.

## FEELER REFLECTION GAME

*The Feeler Reflection Game* is a board game, where the players are invited on a journey by moving from one square to another in turns. In the game, each player picks a predefined persona from six options. The players' personas are all university students, aged 19 to 23, that have different motivations for self-monitoring (Table 1).

| | |
|---|---|
|  | **JENNA** – 23-year-old medical student<br><br>Jenna has suffered from migraines and insomnia since she was a teenager. Throughout time, she has learned to listen to her body in order to prevent the headaches and the insomnia. She likes to experiment and find new ways of doing things. |
|  | **PEKKA** – 19-year-old industrial design student<br><br>Pekka enjoys testing new gadgets and mobile apps. For him, understanding how something works is a challenge he can't let go. Once he has solved a problem, he completely forgets it. He is quite dispersed, and quite often he has problems staying attentive for long periods of time. |
|  | **MING** – 19-year-old computer science student<br><br>There's nothing Ming enjoys more than work that's done well. She is very organized but has a terrible memory. For this reason, she writes lists of things to do, so she can make sure she doesn't miss anything. Her ability for prioritizing is appreciated when leading group projects. |
|  | **SAMIR** – 22-year-old business student<br><br>Samir likes to think that everything has a solution; it's just a matter of having enough data. He also thinks that everyone has the limits one wants to have. He is ambidextrous, and when he was a child he was diagnosed as dyslexic. He is not afraid of mistakes, and he enjoys trying new things. |
|  | **OLGA** – 20-year-old communication student<br><br>Olga makes great use of social media. She has traveled a lot, and she likes to stay in touch with all her friends. Feeling part of a group is very important to her. Over the last two years, she has experienced some tiredness and mild depression during the winter term. |
|  | **TIMO** – 21-year-old art student<br><br>Timo is very social, but sometimes he wants to have some time for himself. From his point of view, one should focus on the present and experience it with all the senses. Even if he makes use of digital technologies, he is quite skeptical about its benefits. |

Table 1. Feeler Reflection Game persona descriptions
Source: Feeler Reflection Game

The goal of the game is to change the behavior of the persona each player is playing in order to improve the personas' learning performance. For this, players need to reach the last square of the game board. Although the game is not competitive, all players should go through all the squares of the board to earn a minimum of eighteen points, which will enable them to move to the final square. The game has ten squares, each one requiring that players perform a specific action. The actions in the squares are:

Square 1: Choose a persona. Decisions are made based on the image displayed in the pawn. Once the selection is made, players receive the character card corresponding to their persona and have some time for carefully examining the profile information.

Square 2: Define the persona degree of motivation for engaging in self-monitoring. On each of the character cards, the persona's level of motivation for improving their well-being is expressed quantitatively. On this square, participants have the opportunity to increase the number initially given by rolling the dice and adding the points earned.

Square 3: Select the type of data to monitor. Players choose from a pre-made list that includes monitoring: a) learning performance and sleep, b) learning performance and physical activity, and c) learning performance, sleep, and physical activity.

Square 4: Define a research question. Participants need to write a concrete question that they can answer through the data. The aim is to gain a better understanding of the persona's condition and, therefore, to be able to find a solution.

Square 5: Pick a device for data collection. Players can choose between the following options for gathering their data: a) by hand, b) mobile app, c) mobile in-built sensor, or d) wearable device. Each device is numerically evaluated according to several features dealing with battery, data management, comfort, cost, and data accuracy. All devices award the same amount of points, but are differently distributed depending on the tool strengths and weaknesses. Players have to make their choice, taking into consideration their character's idiosyncrasy. On this square, participants receive a sticker containing the information about the data collection device chosen that they have to attach to their character's card.

Square 6: Collect the data. In this square, players must choose between two card piles: one dedicated to challenges dealing with data collection, whereas the other one is dedicated to positive aspects. Players roll the dice and, depending if the result is even or odd, they take a card from a different pile. If the player gets a challenge card dealing with a particular feature, some points are subtracted from the total amount assigned to the device. In the opposite case, if the player takes a card focused on the advantages, some points are added.

Square 7: Choose a data visualization type. Participants can visualize their data using: a) a bubble chart, b) a line chart, c) a radar chart, d) a metaphor, or e) a dashboard. Once they have made their selection, they get a sticker with the representation of the data visualization type.

Square 8: Analyze the data. Again, on this square, there are two card piles. In one pile, cards refer to positive aspects of data visualization and give points. In the other pile, cards are focused on data visualization challenges. If players take one of those cards, they get zero or negative points. Players decide from which pile they can take the card by rolling the dice.

Square 9: Determine the readiness for a change. Players count the total amount of points earned during the game in order to determine if they can move to the next square. Those who get more than eighteen points are supposed to have enough motivation, data, and insights to modify their habits and, therefore, reach the last square. Those who haven't gained the necessary amount of points to access the last square must remain in this position.

Square 10: End of the game. Players who reach this square have successfully completed the game.

The game pieces include:

- Pawns with a printed representation of the characters available for players.




Figure 1. Images of the design game used during the contextual inquiry of the Feeler prototype.


http://de2014.uniandes.edu.co | October, 2014. ISBN 978-958-774-070-7. pp. 349-356

- Character cards containing the personas' descriptions, in which players can annotate their game choices.
- Stickers of the devices for data collection and the data visualization types.
- Data collection and data visualization cards identifying challenges and possibilities.
- Dice. Although players are expected to consciously make some decisions during the game, on some occasions, they just roll the dice and this determines the consequences.

In the *Feeler Reflection Game*, each pawn is different and is associated with a character card that includes a description, motivation level to solve the particular problem he/she faces, as well as some empty fields that players need to fill in as the game advances (figure 2). The characters portrayed in the game are personas built according to the research developed by Gimpel et al. (2013), as well as to interviews made with potential end users.

The aesthetics of the *Feeler Reflection Game* borrow from traditional game boards. The selected colors and materials seek to recreate a playful, enjoyable environment that arouses players' curiosity. From the beginning, it was considered important to make participants feel confident and willing to experiment. The use of vivid colors, along with simple and roundish shapes, creates a safe environment, which is visually appealing. In order to stress the storytelling side, it was decided not to use photographs and, instead, adopt an aesthetic based on hand-drawing for the characters and icons for the rest of the game materials.

## ANALYSIS OF FEELER REFLECTION GAME INTERACTIONS

The *Feeler reflection Game* has been created in the framework of a design research looking for solutions to help people to reflect on their well-being and learning. The game is used to elaborate on a design concept called Feeler, a software and service that is able to visualize learning performance and well-being based on data collected from online learning services and tools capturing physical activities, such as movement, exercise, nutrition, and sleeping. The aim of *Feeler* is to encourage people to reflect on their lifestyle and its impact on their learning capabilities.

The *Feeler reflection game* has been developed for a series of workshops of the *Feeler* prototype. In the following, we focus on the latest, which was conducted with five graduate students of art and design. The three design research objectives of the game and playing it with various participants were to:

1. Develop empathic understanding of participants' practices and needs related to self-monitoring,
2. Share and explore the design challenges with the participants related to reflection and behavior change,
3. Obtain feedback and validate initial design concepts that are based on previous research.

Participants started the workshop by playing the *Feeler reflection game*, lasting approximately 45 minutes. The game was followed by a discussion about several issues dealing with the prototype design, such as motivation, data collection, data visualization, and how to better support behavior change. The data gathered during the game for further analysis included a video recording of the session, participants' character cards on which they annotate their game actions, and the design researcher's notes.

In order to infer to what extent playing a game helped participants develop an empathic understanding of the design goals and challenges, qualitative content analysis was used as a method for analyzing the data. Qualitative content analysis is, according to Sandelowski (2000, p.338), 'a dynamic form of analysis of verbal and visual data that is oriented toward summarizing the informational contents of that data'. The codes used for categorizing the data were generated from the data in the course of the study. In this case, the codes that show an empathetic understanding of the situations presented are identification, role-playing, connection with real-life experiences, and creation of a relaxed atmosphere.

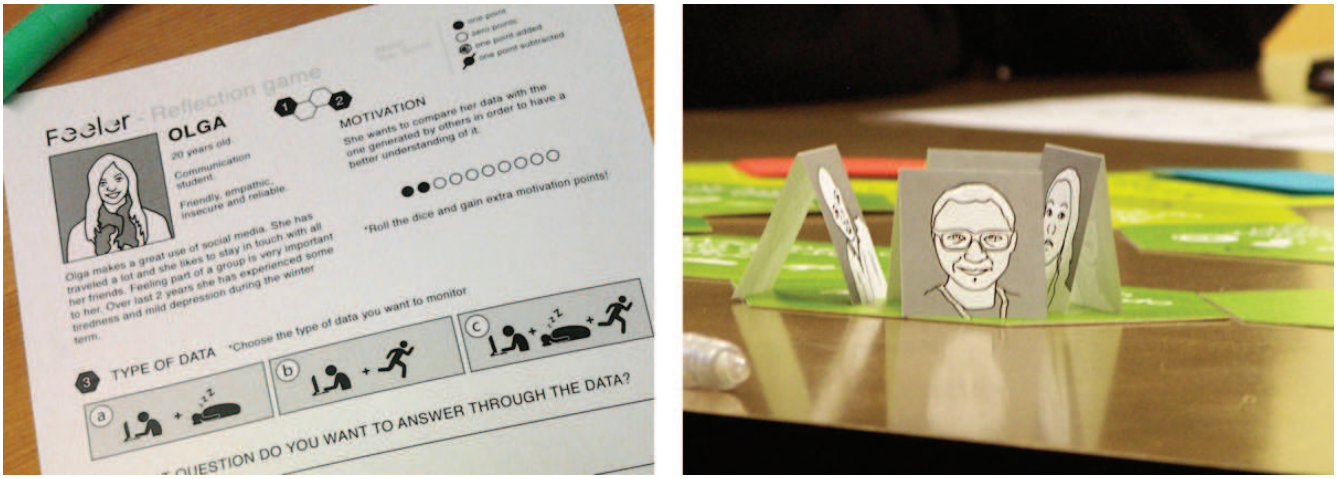



Figure 2. Images of the *Feeler Reflection Game* character cards and pawns.

At the beginning of the game, players picked their character based on the pawns' images. Despite the lack of information, participants' choices showed some sort of identification with the character, mainly based on physical resemblance.

> Player 3: 'I... uhmmm.... Actually I want to be this one (referring to the character). My hair use to be like that'.

Transcript 1. Player picks the character she would use during the game.

After the character selection, players received a characters' card in which they could read a short description of their characters' traits and motivation for engaging in self-monitoring. This background information helped them to make some decisions and to engage in role-playing.

> Player 1: 'I want to get feedback if what I'm doing is good for my health... because I have some mild depressions in winter... and also I feel quite tired so I want to monitor my sleep to see if I can find some relationship between... I don't know yet'.

Transcript 2. Player tries to define a research question to answer through self-monitoring.

One aspect that indicates the players' identification with their characters is the use of first person when referring to their game persona. In some cases, players referred to their character using first person at all times, therefore, showing a high level of involvement in the role-play; whereas in others, they only used third person or alternated between both forms.

> Player 5: 'Pekka has this problem of keeping attention, so then, how long I sleep, but also how I... So, whether if my physical activity and my sleep impact my attention span'.

Transcript 3. Player combines the use of third and first person when talking about her game character.

Another aspect that shows players' empathy towards their characters deals with the connection between real-life and game decisions. In this regard, it is interesting to note that, before playing the *Feeler Reflection Game*, some of the players had experimented with self-monitoring of different aspects of their habits. Curiously, in the game, rather than experimenting with new ways of self-monitoring, some of them replicated what they had tried in real-life.

> Player 2: 'Yes it was. In my case, I was monitoring sleep in the same way I was acting in the game'.

Transcript 4. Player recognizes she self-monitored herself in the same way as in the game.

Despite that self-identification with the characters only happened while playing the game, we can infer that it allowed for an empathic understanding of the personas and scenarios dealing with self-monitoring. From our point of view, role-playing and self-identification contributed to a meaningful and situated discussion about the design challenges of self-monitoring tools.

The use of a game allowed combining formal and informal registers. This was noticeable in the case of the facilitator, who used a more formal register when explaining about the research aims and the instructions of the game and a more informal one while playing. During the game, the use of storytelling techniques enabled the facilitator to abandon the academic register and get into the situation by making jokes, naming players by their characters' names and thus creating a playful and relaxed atmosphere. In this regard, the use of humor contributed to liberate players from prejudices and spontaneously express themselves.



Figure 3. Workshop participants playing the *Feeler Reflection Game*.

Despite the fact that the game board presented a linear timeline highlighting main moments in the process of self-monitoring, participants felt confident enough to bring other topics to the discussion, which were not initially foreseen, such as reward systems or other types of data they would like to monitor, such as stress, weather conditions, or nutrition.

At some point, they also considered it necessary to rearrange the order of the board squares, since, from their point of view, some decisions, such as the definition of the research question, should be approached at the very beginning of the process. The fact that each square on the board was independent from the others enabled the rearrangement of the squares to be fast and easily. In this regard, the visualization of information, either by noting down information on the characters' cards, or by altering the initially given order of the squares, contributed to maintaining the focus and discussing the issues in detail.

During the discussion, participants reflected about the use of a game as a way for presenting academic research. Although they enjoyed playing the *Feeler Reflection Game*, from their


http://de2014.uniandes.edu.co | October, 2014. ISBN 978-958-774-070-7. pp. 349-356

point of view, it was necessary to improve the game playability. According to the participants' comments, the game was too short and offered few opportunities to make decisions that had consequences later on. Since the game was intended to be used at the beginning of the workshop as a way to present the research topics, it was considered important to keep a time limit. However, according to participants, these initial constraints are not so important, and stress should be placed on creating opportunities for players to meaningfully intervene.

## DISCUSSION

*The Feeler Reflection Game* was designed during the contextual inquiry of the *Feeler* work-in-progress prototype. According to Leinonen (2008, 2010), contextual inquiry is an initial stage of the research-based design process that consists in an exploration of the socio-cultural context of the design. During the contextual inquiry, designers may use rapid ethnographic methods, such as participant observation and user interviews, for exploring design situations and defining design challenges. Although design games have been highly used in co-design sessions in PD (Brandt 2006; Vaajakallio, 2012), we consider that they are powerful tools to use in the early stages of the design process in order to create a productive dialogue (Kronqvist et al., 2012; Vaajakallio, 2012).

Considering that the *Feeler Reflection Game* was used as a strategy for presenting end users' different personas and initial scenarios of the use of the future design, it shares some characteristics with the scenario-oriented design games defined by Brandt (2006). According to this author, 'enacted scenario construction can be viewed as an exploratory design game because it involves a play with props, takes place within a pre-defined location, is limited in time, and follows specific rules' (2006, p.59). In the *Feeler Reflection Game*, the creation of personas allowed participants to interpret a role and develop it further according to their own views. Similarly to the Brandt et al. *User Game* (2008), the *Feeler Reflection Game* aimed to create a shared image, between designers and workshop participants, of the end users, based on the field data.

The generation of scenarios in design is a well-known practice. According to Schön (1983), the creation of scenarios is a design technique that helps to restructure a design situation and provides new insights. The scenarios presented in the *Feeler Reflection Game* described a particular interpretation of a QS tool use situation, but incomplete and open to negotiation (Carroll, 2000). The actions taken and decisions made by players when playing the game contributed to recreating and completing the scenario of use.

> Player 1: 'I could do the same thing (noting down the data by hand as another player decided to do), but I'm a computer science student (referring to the persona she picked), so I use technology (a smart phone with an built-in sensor)'.

Transcript 5. Player explains why her character makes a certain decision.

The game board layout offered participants a comfortable environment in which they could already advance certain aspects, such as the presence of rules and the existence of a certain path, among others (Johansson, 2005). Furthermore, as Kronqvist et al. (2012) highlight, the game board acted as a design medium with a double function: presenting information to participants and engaging them to freely communicate personal experiences and views. The game board was also a great resource for guiding and informing the discussion about the concept design of the future prototype.

The game materials were used as 'things-to-think-with' (Papert, 1980; Brandt, 2006) that participants used to analyze the situation and give new meanings to it. The game offered a common language that helped structuring the argumentation and exchange of people's views (Brandt & Messeter, 2004). In this regard, as Brandt (2006) and Kronqvist et al. (2012) outline, game materials can function as boundary objects (Star, 1989), since they support dialogue between different disciplines and interests.

Finally, we would like to present the design of games as a design work (Johansson, 2005). The construction of a design game requires designers to structure and define what would be at the center of the design process. In the case of the *Feeler Reflection Game*, a lot of effort was put into presenting previous research results in a suggestive and playful way, so participants would feel comfortable to freely express themselves and contribute to the concept design. In this regard, the game design had its own process. A couple of weeks before the workshop, a low-fi prototype of the game was tested, and some aspects were redesigned according to the feedback provided. The experience of creating a game was an interesting exercise for enabling designers to identify with end users and for improving communication techniques with them.

## CONCLUSIONS

In this article, we present design games as a useful strategy in PD to develop an empathic understanding of end users from early stages of the design, concretely during the contextual inquiry. Design games act as boundary objects that provide an engaging and pleasurable experience that facilitates players' free expression, and in which participants' concerns are taken into consideration.

According to our experience, design games work in two different ways. In the case of users, the game players, the game offers a unique environment for presenting previous research results, such as scenarios and personas through storytelling techniques. This way, players can easily identify themselves with the personas and scenarios and contribute to the research by helping to envision future practices in design.

From our perspective, as designers and researchers, creating a design game requires putting ourselves in the end users' position in order to present a situation that stimulates end users' communicability and creativity. In this regard, the game works as an example of empathic design since it seeks to find

inspiration in end users by empathizing with them. Furthermore, the game works as workshop agenda, since it contributes to presenting the design topics and creating an experience on which players can discuss later. Considering all the above-stated information, design games offer great opportunities for developing mutual learning and empathy among end users and designers.

## REFERENCES


Binder, T. (1999) 'Setting the stage for improvised video scenarios', *CHI'99 extended abstracts on Human factors in computing systems*, (May), pp. 230-231.

Botero, A. (2013) *Expanding Design Space(s). Design in Communal Endeavours*. Helsinki: Aalto ARTS Books.

Bannon, L. J., and Ehn, P. (2012) 'Design Matters in Participatory Design', in (Simonsen and Robertson*Routledge Handbook of Participatory Design*. New York: Routledge, pp.37-63.

Brandt, E. (2006) 'Designing exploratory design games: a framework for participation in Participatory Design?', *Proceedings of the ninth conference on Participatory design: Expanding boundaries in design*, 1, (August), pp.57-66.

Brandt, E. and Grunnet, C. (2000) 'Evoking the future: drama and props in user centered design', *Proceedings of the Participatory Design Conference*, place of conference, CPSR.

Brandt, E., and Messeter, J. (2004) 'Facilitating collaboration through design games' *Proceedings of the eighth conference on Participatory design: Artful integration: interweaving media, materials and practices*, 1, (July), pp.121-131.

Buchenau, M., and Suri, J. F. (2000) 'Experience prototyping', *Proceedings of the 3rd conference on Designing interactive systems: processes, practices, methods, and techniques*, August, pp. 424-433.

Carroll, J. (2000) *Making Use: Scenario-Based Design of Human-Computer Interactions*. Place of publication: MIT Press.

Costikyan, G. (2000) 'Where stories end and games begin', *Game Developer*, 7, (9), pp.44-53.

Ehn, P. (1988) *Work-oriented design of computer artifacts*. Stockholm, Sweden: Arbetslivscentrum.

Ehn, P., and Sjøgren, D. (1991) 'From system descriptions to scripts for action', *Design at work: Cooperative design of computer systems*, volume/issue no., pp.241-268.

Fulton Suri, J. (2003) 'Empathic design: informed and inspired by other people's experience', in Koskinen, I., Battarbee, K., and Mattelmäki, T. (eds.) *Empathic Design*. Finland: IT Press, pp.51–57.

Gaver, W., Dunne, T. and Pacenti, E. (1999) 'Cultural Probes', *Interactions*, 6, (1), pp.21–29.

Gimpel, H., Nißen, M., and Görlitz, R. (2013) 'Quantifying the Quantified Self: A Study on the Motivations of Patients to Track Their Own Health', *Proceedings of 2013 International Conference on Information Systems* (ICIS 2013), 15-18 December, 2013, Milan, Italy.

Habraken, N. J., and Gross, M. D. (1988) 'Concept design games', *Design Studies*, 9, (3), pp.150-158.

Johansson, M. (2005) *Participatory Inquiry – Collaborative Design*. School of Engineering. Blekinge Institute of Technology, PhD dissertation.

Koskinen, I., Battarbee, K. and Mattelmäki, T. (2003) *Empathic design, user experience in product design*. Helsinki: IT Press.

Kouprie, M. and Visser, F. S. (2009) 'A framework for empathy in design: stepping into and out of the user's life', *Journal of Engineering Design*, 20, (5), pp.437-448.

Kronqvist, J., Järvinen, M. and Leinonen, T. (2012) 'Games as Design Medium. Utilizing Game Boards for Design Inquiry with Cancer Patients' *Proceedings of ServDes.2012 – Service Design and Innovation Conference*, 8-10 February 2012, Espoo, Finland.

Leinonen, T., Toikkanen, T., and Silfvast, K. (2008) 'Software as hypothesis: research-based design methodology', *Proceedings of the Tenth Anniversary Conference on Participatory Design 2008*, pp.61-70, Indiana University.

Leinonen, T. (2010) *Designing learning tools - methodological insights*. PhD Dissertation, Aalto University. Jyvaeskylae: Bookwell.

Mattelmäki, T. (2006) *Design Probes*. PhD Dissertation, University of Art and Design Helsinki, Publication series A 69, Finland.

Muller, M. J. (2003) 'Participatory design: the third space in HCI', *Human-computer interaction: Development process*, volme/issue no., pp.165-185.

Papert, S. (1980) *Mindstorms – Children, Computers and Powerful Ideas*. New York: Basic Books Inc. Publishers.

Salen, K. and Zimmerman, E. (2004) *Rules of Play – game design fundamentals*. England: The MIT Press.

Sandelowski, M. (2000) 'Focus on research methods-whatever happened to qualitative description?', *Research in nursing and health*, 23, (4), pp.334-340.

Sanders, E. B.-N. and Dandavate, U. (1999) 'Design for experiencing: New tools', *Proceedings of the First International Conference on Design and Emotion*. Delft University of Technology, Delft, Netherlands, pp.87–92.

Schön, D. A. (1983) *The reflective practitioner: How professionals think in action* (Vol. 5126). London: TempleSmith..

Steen, M., Kuijt-Evers, L. and Klok, J. (2007) 'Early user involvement in research and design projects – A review of methods and practices', *23rd EGOS Colloquium*, Vienna.

Star, L. S. (1989) 'The structure of Ill-structures Solutions: Heterogeneous Problem-Solving, Boundary Objects and Distributed Artificial Intelligence', *Distributed Artificial Intelligence*, 2, pp.37-54.

Vaajakallio, K. (2012) *Design games as a tool, a mindset and a structure*. PhD Dissertation, Aalto University, School of Arts, Design and Architecture.



http://de2014.uniandes.edu.co | October, 2014. ISBN 978-958-774-070-7. pp. 349-356